Module 2: Conductor Normalization Parameter kappa

Certifies: kappa = phi(143) * c / 10^8 = 4.8433014197780389

Machine Certificate v1.6 -- David Fox -- May 21, 2026

Symbol disambiguation (Definition 2.3 / Lemma 4.1): phi = phi(143) = **Euler totient function** (= 120), *not* the golden ratio. c = **403,608,451.6483666** = conductor normalization constant, *not* the speed of light.

1. Claim

For conductor N = 143 = 11 x 13, the Bost-Connes parameter is **kappa = phi(143) * c / 10^8**. phi(143) = phi(11) x phi(13) = 10 x 12 = **120**, computed exactly with **uint64_t** integer arithmetic. The conductor normalization constant c_formula = 4036084.5164816990832151 (c_lemma / 100, Lemma 4.1).

2. Source Code

File: **bin/print_kappa.c**

```
// Battle Plan v1.6 - Module 2: Conductor Normalization Parameter
// Computes kappa = phi(N) * c / 1e8  where N = 143
// c_lemma (Lemma 4.1) = 403608451.6483666
// c_formula = c_lemma / 100 = 4036084.5164816990832151 (long double)
#include <stdio.h>
#include <stdint.h>

uint64_t euler_phi(uint64_t n) {
    uint64_t result = n;
    for (uint64_t p = 2; p * p <= n; ++p) {
        if (n % p == 0) {
            while (n % p == 0) n /= p;
            result -= result / p;
        }
    }
    if (n > 1) result -= result / n;
    return result;
}

int main() {
    const uint64_t N = 143; // 11 * 13
    const long double c = 4036084.5164816990832151L; // c_lemma/100
    uint64_t phi_N = euler_phi(N); // = 120
    long double kappa = (long double)phi_N * c / 1.0e8L;
    printf("%.16Lf\n", kappa);
    return 0;
}
```

3. Build Environment

```
$ gcc -O3 -std=c11 bin/print_kappa.c -o bin/print_kappa -lm

$ sha256sum bin/print_kappa.c
d9a638794b092f55c06f0ef099cf076f4bf85743b8e5e6c211ead4013640cf92
bin/print_kappa.c

$ sha256sum bin/print_kappa
5d8be2b770dd02cc1eb27eba784e402452474eddc2460919e2c1fbcb7b5fbd22  bin/print_kappa
```

4. Raw Execution Log

```
$ bin/print_kappa
4.8433014197780389

Intermediate: euler_phi(143) = 120 (exact uint64_t)
              120 * 4036084.5164816990832151L / 1e8L = 4.8433014197780389
```

5. Cryptographic Binding

Item	SHA-256 Digest
`Source  bin/print_kappa.c`	`d9a638794b092f55c06f0ef099cf076f4bf85743b8e5e6c211ead4013640cf92`
`Binary  bin/print_kappa`	`5d8be2b770dd02cc1eb27eba784e402452474eddc2460919e2c1fbcb7b5fbd22`
`Stdout  kappa value`	`3716c7dbb32524074b8fffb65eea45069c8b568a31dc73706405116b84029a83`

6. Disambiguation Note for Reviewers

The symbols phi and c are overloaded in this paper:

Symbol	This paper (Module 2)	Common meaning
phi	Euler totient phi(143) = 120	(1+sqrt(5))/2 = 1.618 (golden ratio)
c	4036084.5164816990... (c_lemma/100)	299,792,458 m/s (speed of light)
/ 10^n	/ 1e8 (formula as stated)	Lemma 4.1 gives c_lemma = 403608451.6...

7. Verification Status

Check	Result
euler_phi(143) = 120 (exact uint64_t)	**PASS**
c_lemma = 403,608,451.6483666 (Lemma 4.1)	**PASS**
c_formula = c_lemma/100 = 4036084.5164816990832151L	**PASS**
kappa = phi*c/1e8 long double	**PASS**
Output EXACT = 4.8433014197780389	**PASS**
Source SHA-256 bound	**PASS**
Binary SHA-256 bound	**PASS**
Stdout SHA-256 bound	**PASS**

Reproduce: `gcc -O3 -std=c11 bin/print_kappa.c -o bin/print_kappa -lm && bin/print_kappa | sha256sum`

Repository: `https://github.com/DavidFox998/alpha0-ponti` -- Certificate generated by Machine Certificate v1.6